The impact of tropospheric versus stratospheric CO_2 forcing on the Hadley cell

Climate dynamics department meeting

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DER FORSCHUNG | DER LEHRE | DER BILDUNG

The Hadely cell (HC) under greenhouse gas forcing

Model intercomparison studies

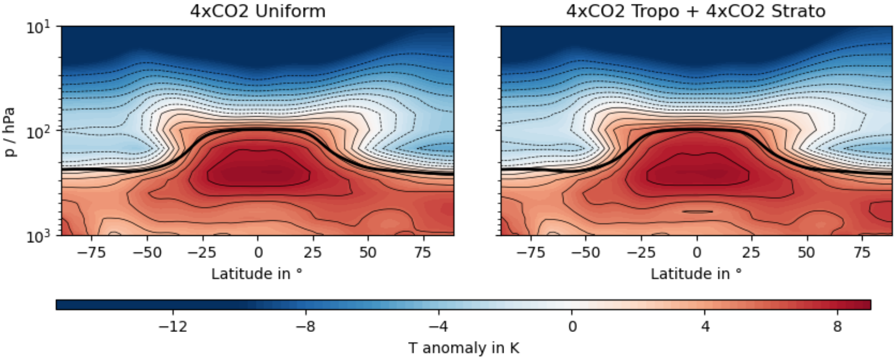
- HC **weakening** of 1.2%/K [Lu et al., 2007]
- HC **widening** at the poleward edge of 0.7°/K [D'Agostino et al., 2017]
- Linked to changes in tropopheric depth, meridional temperature gradient, and vertical stability [Lionello et al., 2024]

Is this also true if we decompose the CO_2 forcing vertically.

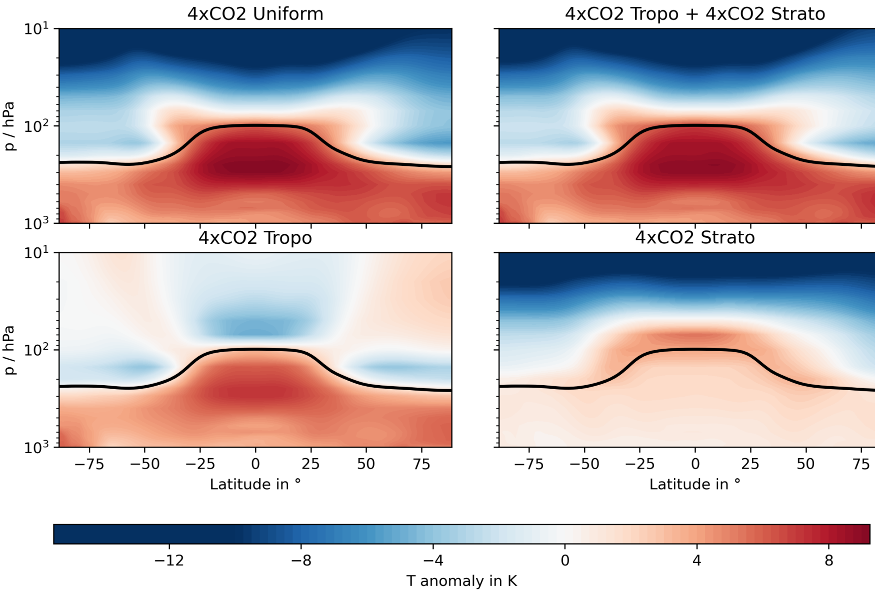
Model: ECHAM 6

- CMIP6 type general circulation model
- Mixed-layer ocean ($d = 50$ m) configuration
- 47 vertical pressure levels with (lat,lon)$= (96, 192)$ gridpoints (~ 200 km)
- 4 runs: pre-industrial, uniform 4xCO_2, 4xCO_2 troposphere, 4xCO_2 stratosphere

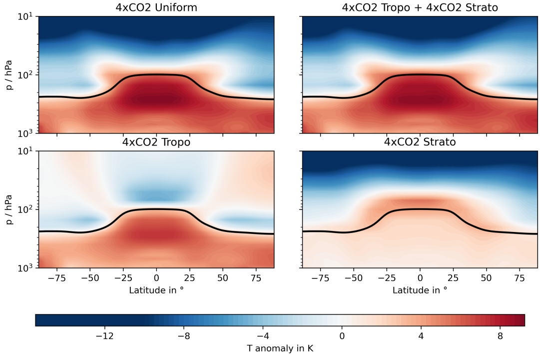
Linear decomposition of forcing works



Temperature anomalies



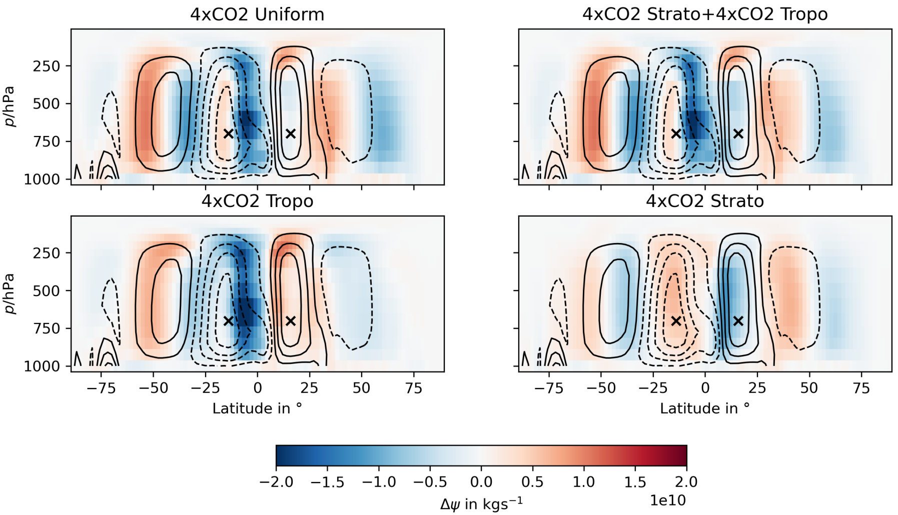
Temperature anomalies



Yet to be clarified

- Stratospheric cooling in 4xCO_2 Tropo
- Stratospheric warming in 4xCO_2 Strato
- one-dimensional RCE model *konrad*

Zonally averaged circulation



Changes in Hadley circulation

	Uniform	Tropo	Strato
$\Delta\psi_{\mathrm{max,NH}}$	-1.6%	3.4%	-8.7%
$\Delta\psi_{\mathrm{max,SH}}$	-1.4%	5.6%	-5.7%
$\Delta\Phi_{\mathrm{H,NH}}$	1.6°	0.8°	0.8°
$\Delta\Phi_{\mathrm{H,SH}}$	-1.6°	-1.0°	-0.6°

Surprise

- HC is **weakening** with stratospheric CO_2
- HC is **strengthening** with tropospheric CO_2

Thesis topic

Research gap

Why is the HC weakening with stratospheric CO_2 forcing but strengthening with tropospheric CO_2 forcing.

Thesis approach

Employ dynamical theories of the HC to explain this behavior.

The axisymmetric theory of the HC

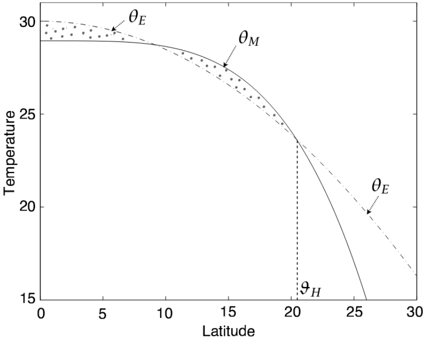


Figure: The edge of the HC [Vallis, 2017].

Width and extent of the HC [Held and Hou, 1980]

$$\Phi_H \propto (H\Delta_h)^{1/2} \quad (1)$$

$$\psi_{\max} \propto \frac{(H\Delta_h)^{5/2}}{\Delta_v} \quad (2)$$

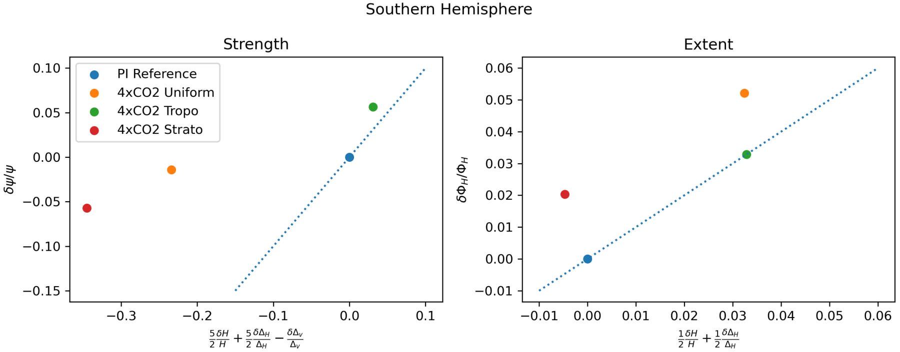
H: Troposphere depth
Δ_h: Fractional pole to equator temperature difference
Δ_v: Fractional surface to tropopause temperature difference

Linearisation of direct thermodynamical drivers

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2}\frac{\delta H}{H} + \frac{5}{2}\frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v}$$
$$\frac{\delta\Phi_{\mathrm{H}}}{\Phi_{\mathrm{H}}} = \frac{1}{2}\frac{\delta H}{H} + \frac{1}{2}\frac{\delta\Delta_h}{\Delta_h}$$

- meridional near-surface potential temperature contrast $\Delta_h = \bar{\theta}(800 - 700\ \mathrm{hPa}, 0 - 10\ ^\circ) - \bar{\theta}(800 - 700\ \mathrm{hPa}, 80 - 90\ ^\circ)$
- subtropical static stability $\Delta_v = \bar{\theta}(200\ \mathrm{hPa}, 20 - 40\ ^\circ) - \bar{\theta}(800\ \mathrm{hPa}, 20 - 40\ ^\circ)$
- tropical tropopause height $H = \bar{H}(0 - 15^\circ)$

Does linearised H&H explain the HC's response?



Why does H&H fail to explain HC dynamics?

Limitations

- Idealized model without oceans, land mass, clouds, etc.
- Eddies and axially asymmetric flow
- Temperature scaling parameters are in radiative equilibrium

Indirect thermodynamical drivers

Eddy momentum flux

$$\psi(1 - \mathrm{Ro}) \approx \frac{S}{f} \tag{3}$$

- $\mathrm{Ro} \to 1$: Axisymmetric limit
- $\mathrm{Ro} \to 0$: Eddy momentum flux dominates

$\mathrm{Ro} = \xi / f$: Local Rossby number
S: Eddy momentum flux divergence

Intermediate regimes

Eddy momentum flux

$$\psi(1 - \mathrm{Ro}) \approx \frac{S}{f}$$

- $\mathrm{Ro} \to 1$: Axisymmetric limit
- $\mathrm{Ro} \to 0$: Eddy momentum flux dominates

$0 < \mathrm{Ro} < 1$

There is no theory that captures how the Hadley circulation responds to climate changes in this intermediate range of Rossby numbers. - [Levine and Schneider, 2011]

Annual mean and equinox HC are similar

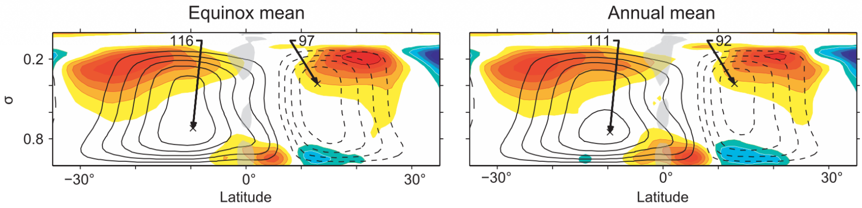


Figure: Local Rossby numbers in equinox and annual mean case using reanalysis data from [Levine and Schneider, 2011].

HC strength and eddy momentum fluxes

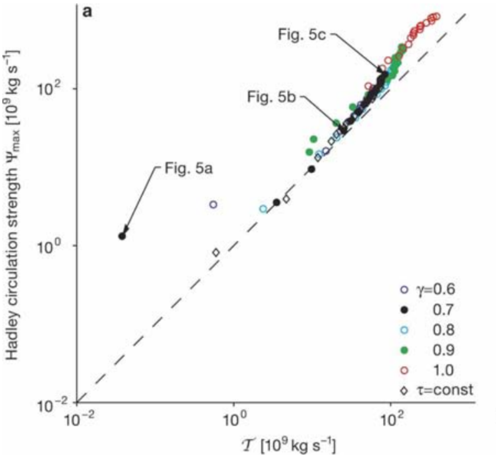
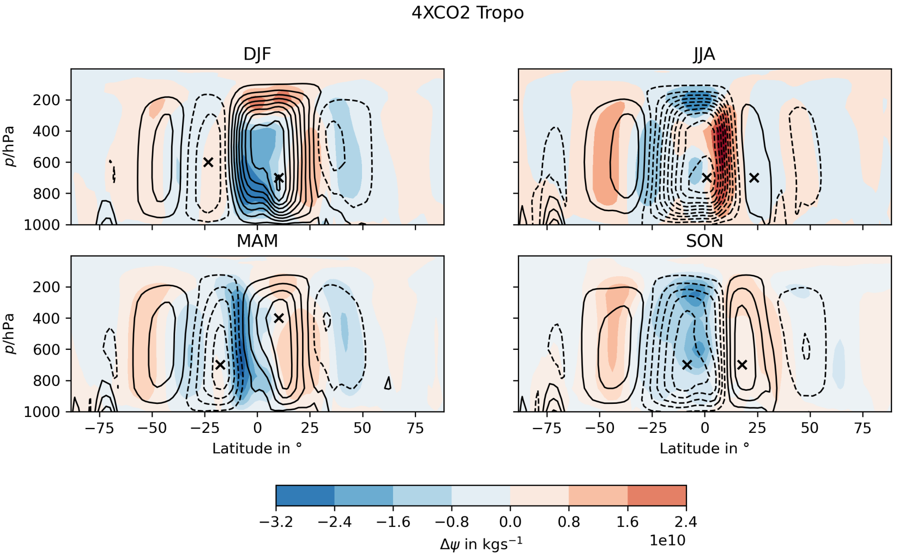


Figure: Linear relation between eddy momentum flux and HC strength from [Walker and Schneider, 2006].

Winter HC is close to angular momentum conserving limit



More frameworks explaining HC dynamics

- Weakening of the overall tropical circulation [Vecchi and Soden, 2007]
- Energy budget constraints and GMS [Chou and Chen, 2010]

Summary

Research gap

We have found the annual mean HC to strengthen under tropospheric CO_2 forcing while weakening with stratospheric CO_2 forcing.

Thesis approach

Simplified, axisymmetric frameworks (Held&Hou) cannot explain this response. Indirect thermodynamical drivers will be investigated in a next step.

Thank you for your attention! Do you have any questions?

References I

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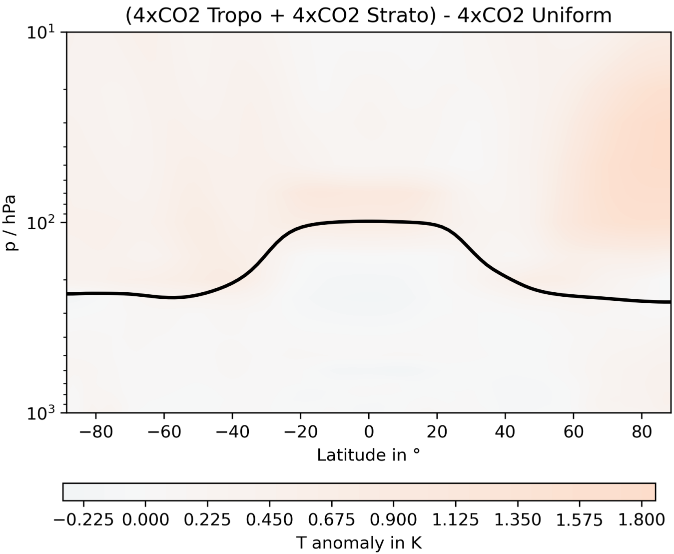
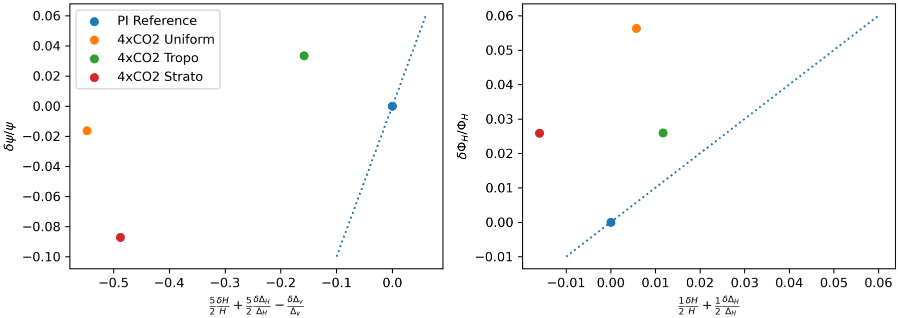
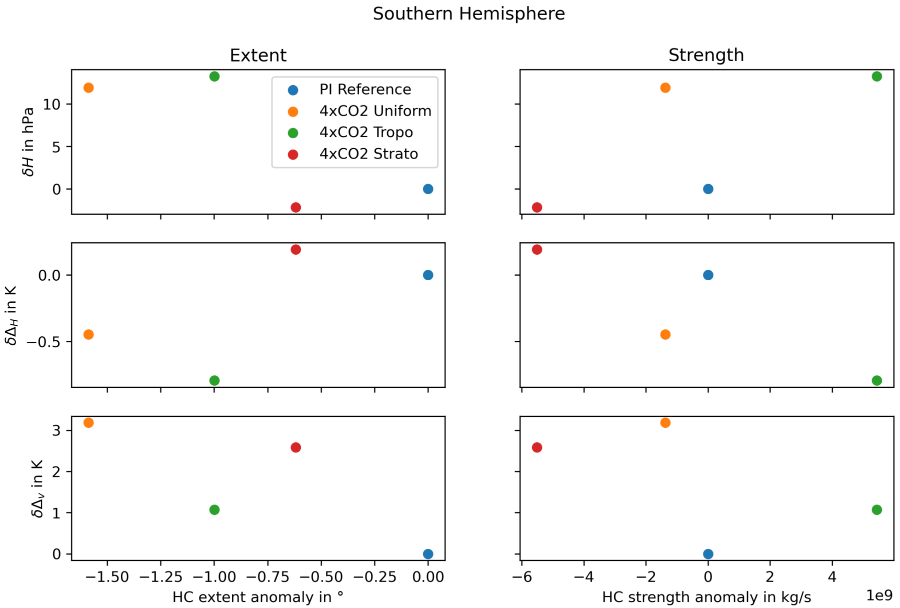
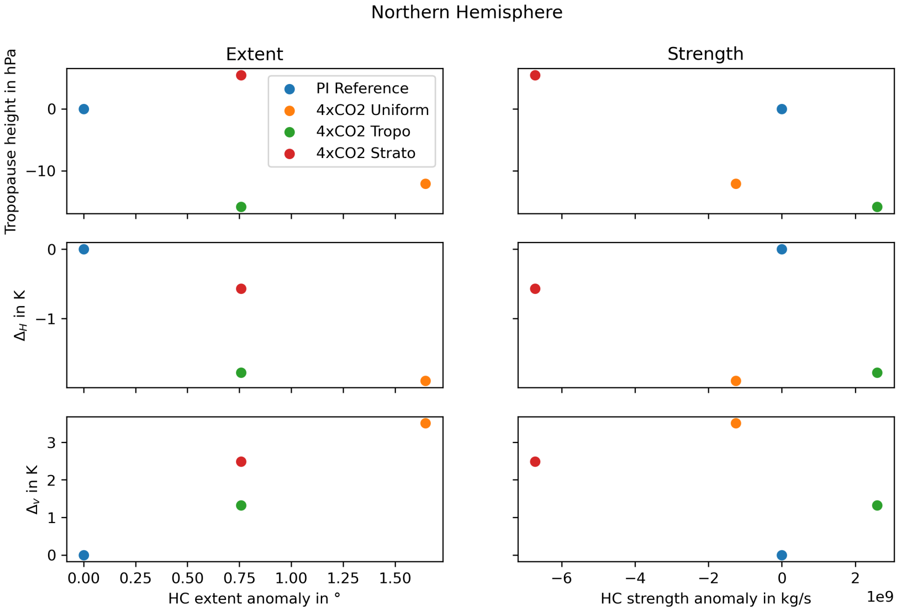
Is the decomposition valid?

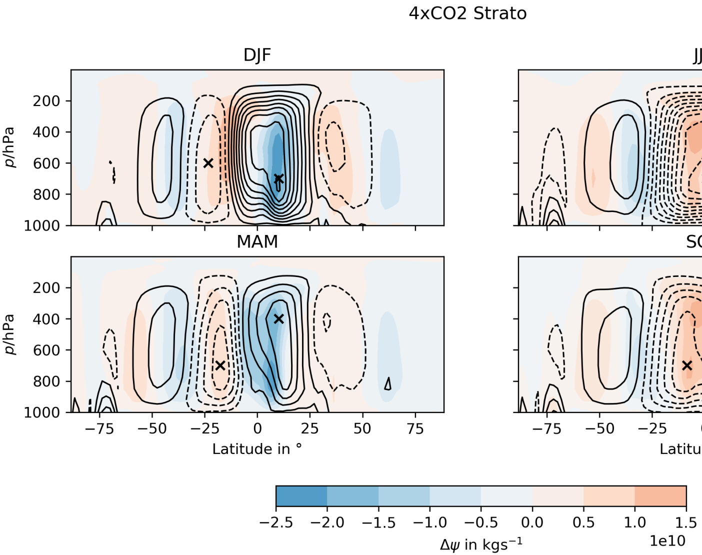


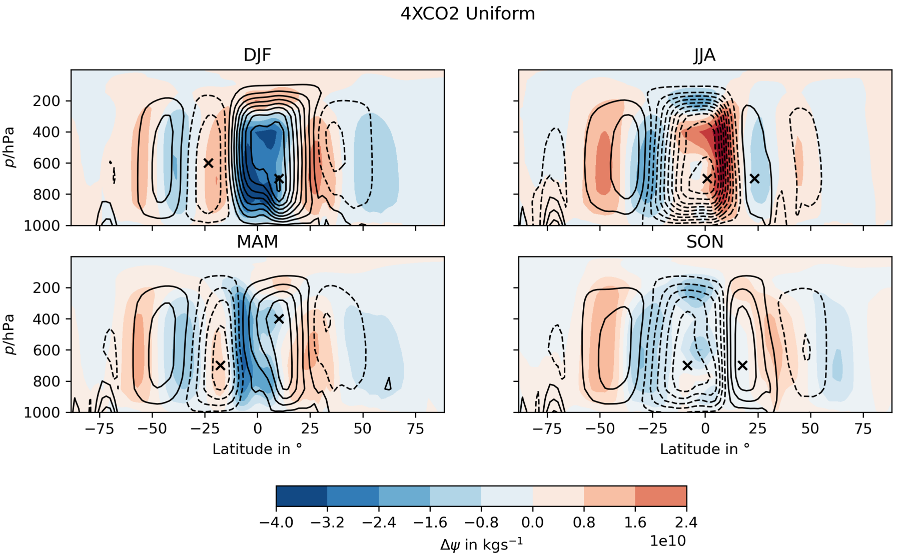
Northern hemisphere




Southern Hemisphere
Extent
Strength
PI Reference
4xCO2 Uniform
4xCO2 Tropo
4xCO2 Strato
δH in hPa
$\delta \Delta_H$ in K
$\delta \Delta_v$ in K
HC extent anomaly in °
HC strength anomaly in kg/s
1e9


Northern Hemisphere
Extent
Strength
PI Reference
4xCO2 Uniform
4xCO2 Tropo
4xCO2 Strato
Tropopause height in hPa
Δ_H in K
Δ_V in K
0
−10
−1
3
2
1
0.00
0.25
0.50
0.75
1.00
1.25
1.50
HC extent anomaly in °
−6
−4
−2
0
2
1e9
HC strength anomaly in kg/s


4xCO2 Strato
DJF
JJA
MAM
SON
p/hPa
200
400
600
800
1000
−75
−50
−25
0
25
50
75
Latitude in °
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Δψ in kgs⁻¹
1e10



4XCO2 Uniform
DJF
JJA
MAM
SON
p/hPa
200
400
600
800
1000
−75
−50
−25
0
25
50
75
Latitude in °
−4.0
−3.2
−2.4
−1.6
−0.8
0.0
0.8
1.6
2.4
Δψ in kgs⁻¹
1e10